4.3 Sine Function and Inverse Sine Function

Let us recall that sine function is a function with $\mathbb{R}$ as its domain and $[-1, 1]$ as its range. We write $y = \sin x$ and $y = \sin^{-1} x$ or $y = \arcsin(x)$ to represent the sine function and the inverse sine function, respectively. Here, the symbol -1 is not an exponent. It denotes the inverse and does not mean the reciprocal.

We know that $\sin(x + 2\pi) = \sin x$ is true for all real numbers x. Also, $\sin(x + p)$ need not be equal to $\sin x$ for $0 < p < 2\pi$ and for all x. Hence, the period of the sine function is 2π.

4.3.1 The graph of sine function

The graph of the sine function is the graph of $y = \sin x$, where x is a real number. Since sine function is periodic with period 2π, the graph of the sine function is repeating the same pattern in each of the intervals, $\cdots$, $[-2\pi, 0]$, $[0, 2\pi]$, $[2\pi, 4\pi]$, $[4\pi, 6\pi]$, $\ldots$. Therefore, it suffices to determine the portion of the graph for $x \in [0, 2\pi]$. Let us construct the following table to identify some known coordinate pairs for the points (x, y) on the graph of $y = \sin x$, $x \in [0, 2\pi]$.

x (in **radian**)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \sin x$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0

It is clear that the graph of $y = \sin x$, $0 \le x \le 2\pi$, begins at the origin. As x increases from 0 to $\frac{\pi}{2}$, the value of $y = \sin x$ increases from 0 to 1. As x increases from $\frac{\pi}{2}$ to π and then to $\frac{3\pi}{2}$, the value of y decreases from 1 to 0 and then to -1. As x increases from $\frac{3\pi}{2}$ to 2π, the value of y increases from -1 to 0. Plot the points listed in the table and connect them with a smooth curve. The portion of the graph is shown in Fig. 4.4.

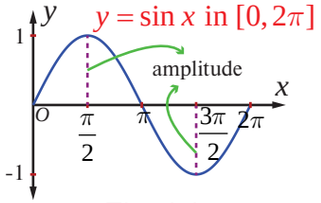


Fig. 4.4

The entire graph of $y = \sin x$, $x \in \mathbb{R}$ consists of repetitions of the above portion on either side of the interval $[0, 2\pi]$ as $y = \sin x$ is periodic with period 2π. The graph of sine function is shown in Fig. 4.5. The portion of the curve corresponding to 0 to 2π is called a **cycle**. Its amplitude is 1.

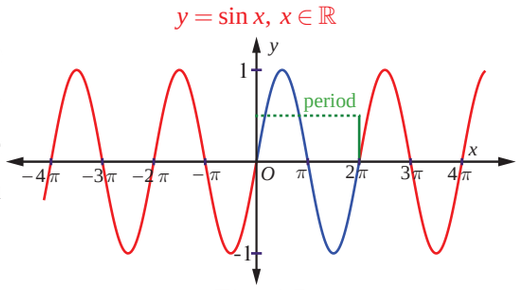


Fig. 4.5

Note

Observe that $\sin x \ge 0$ for $0 \le x \le \pi$, which corresponds to the values of the sine function in quadrants I and II and $\sin x < 0$ for $\pi < x < 2\pi$, which corresponds to the values of the sine function in quadrants III and IV.

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